

Continuous functions – 1

Rigidity: continuity, density of the rationals, pathologies

Homework

1 сентября 2026 года

Theoretical minimum. Continuity in the ε - δ and in the sequential form; oscillation; continuity is determined by the values on a dense set; the intermediate value property; monotone functions and their one-sided limits; continuity together with injectivity forces strict monotonicity; the Cauchy functional equation as the model example of rigidity.

1. (IMC 2007) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Suppose that for every $c > 0$ the graph of f can be moved onto the graph of cf by a translation or a rotation. Does it follow that $f(x) = ax + b$?
2. (Thomae) Let $f(x) = \frac{1}{q}$ for $x = \frac{p}{q}$ in lowest terms and $f(x) = 0$ for irrational x . Prove that f is continuous exactly at the irrational points, and prove that no function on $\mathbb{R}$ is continuous exactly at the rational points. (*Hint: the second half is the Baire category theorem; it is proved in the eighth sheet.*)
3. (IMC 2000) Is it true that if $f: [0, 1] \rightarrow [0, 1]$ is a) monotone increasing, b) monotone decreasing, then there exists $x \in [0, 1]$ with $f(x) = x$?
4. (IMC 2000) Let f be continuous and nowhere monotone on $[0, 1]$. Show that the set of points at which f attains a local minimum is dense in $[0, 1]$.
5. (MIPT 2008) Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous. Prove that there is a subset $X \subseteq [0, 1]$ of cardinality continuum on which f is monotone.
6. (RMO 1995) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be surjective and such that $(f(x_n))$ diverges for every divergent sequence (x_n) . Prove that f is continuous.

7. (RMO 1995) Let $M > 0$ and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $|f(x+y) - f(x) - f(y)| \leq M$ for all x, y . Prove that $\lim_{n \rightarrow \infty} \frac{f(nx)}{n}$ exists for every x ; denoting it by $g(x)$, prove that g is continuous at 0 and deduce that $\lim_{x \rightarrow \infty} \frac{f(x)}{x}$ exists and is finite.
8. (RMO 1997) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be continuous bijections with $(f \circ g^{-1})(x) + (g \circ f^{-1})(x) = 2x$ for all x . Show that if $f(x_0) = g(x_0)$ for some x_0 , then $f = g$.
9. (RMO 1998) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and suppose that for all $a < b$ there are $c_1 \leq c_2$ in $[a, b]$ with $f(c_1) = \min_{[a,b]} f$ and $f(c_2) = \max_{[a,b]} f$. Prove that f is increasing.
10. (RMO 1999) Find all continuous $f: \mathbb{R} \rightarrow [1, \infty)$ for which there are $a \in \mathbb{R}$ and a positive integer k with $f(x)f(2x) \cdots f(nx) \leq an^k$ for all real x and all positive integers n .